

BUILT — Schematic orientation + HD (v0.30.0)

Tilt on BOTH axes, a mirror for working from the opposite side with labels re-drawn readable (text pages), and on-demand HD that renders from the lossless source. All presentation-only; the page data is ne

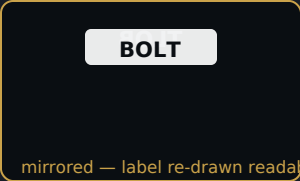
1 · DUAL-AXIS TILT (honest flat-sheet)



rotateY (tilt Y) + rotateX (tilt X)
-60 deg .. +60 deg each

CSS perspective — not reconstructed depth

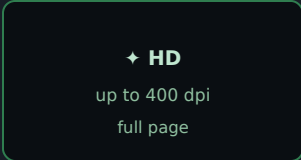
2 · MIRROR + READABLE LABELS



The drawing flips (scaleX -1) so you can orient from the opposite side. On pages with a text layer, each word box (from PyMuPDF, /api/pagewords) is re-drawn un-mirrored at its mirrored position so labels stay readable.

Honest: a mirror is an orientation aid, NOT a true rear view (that needs a different figure). Image-only pages show no overlay until OCR provides boxes.

3 · ON-DEMAND HD



HD re-renders the page from the lossless source PDF at high resolution on demand — full fidelity, no pre-baked duplicates (which would explode storage on the 85GB corpus with no quality gain). The loupe already goes to 700 dpi on the region under the cursor.

Every file keeps its highest fidelity: we render from source, never from a downscaled copy.

INVARIANTS (R1 · R6)

All four controls (tilt X, tilt Y, mirror, HD) are presentation-only and reversible: the page bytes, the index, FTS search, and 104th sheet generation are untouched. Mirror labels are read-only overlays from the PDF text layer; HD changes only render resolution. Nothing is invented — tilt is a flat-sheet rotation, the mirror is a flip, HD is a higher-DPI render of the same source.

NEW / CHANGED

- **Toolbar:** tilt Y + tilt X sliders
each -60..+60 deg, CSS perspective
- **Toolbar:** ↔ Mirror
flip + readable label overlay (text pages)
- **Toolbar:** ↗ HD
full-page render up to 400 dpi from source
- **Server:** /api/pagewords
normalized word boxes from PyMuPDF (OCR-gated)
- **Server:** /page full-page cap 300 -> 400
HD headroom; clip loupe still 700